recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

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Abstract*—Evaluating whether different language-model policy configurations exhibit improved recovery from intermediate reasoning errors is difficult because overall continuation quality can confound recovery-specific behavior. We introduce recovery_eval, a reproducible data-centric evaluation framework that compares verifier-defined recovery states with prospectively matched reference-control states using frozen structural covariates. The framework combines state matching with append-only exposure governance, primitive neural-rollout provenance, and independent analysis reconstruction. We apply recovery_eval to 400 genuine neural continuations generated by two released Qwen2.5-Math 1.5B checkpoint-interface configurations on 20 prospectively isolated GSM8K evaluation problems. The Instruct checkpoint showed higher observed continuation success than the Base checkpoint in both recovery and matched-control states, but the improvement was larger for controls. The resulting matched recovery-specific contrast was minus 0.110, with a descriptive 95 percent problem-level bootstrap interval from minus 0.240 to 0.030. Thus, under this evaluation, we did not observe evidence of a recovery-specific advantage for the Instruct checkpoint. The result illustrates how aggregate benchmark improvements can obscure state-specific reasoning behavior and motivates provenance-aware, state-matched evaluation for language-model diagnostics.

***Index Terms*—Large Language Models, Mathematical Reasoning, Model Evaluation, Data-Centric AI, Reproducibility**

I. Introduction

Large language models (LLMs) trained on multi-step mathematical reasoning demonstrate remarkable capacity for generating complex chain-of-thought solutions [1], [2]. A foundational question in AI evaluation is whether post-training procedures—such as instruction tuning, process supervision, or reinforcement learning from verifier feedback—instill structural reasoning intelligence that enables models to detect and recover from early intermediate errors [3], [4].

Evaluating error-recovery capability is fraught with statistical confounding. Standard benchmark evaluation measures aggregate end-to-end accuracy, which conflates baseline generation fluency with true error recovery. Simply comparing model rollouts from error-containing prefixes against valid prefixes ignores critical confounding covariates, including trajectory depth, remaining solution complexity, problem difficulty, and token length. A post-trained model may exhibit higher accuracy on error prefixes simply because its overall baseline generation quality has improved across all prompt conditions, rather than possessing a specialized error-recovery mechanism.

To address these diagnostic challenges, we introduce recovery_eval, a data-centric evaluation infrastructure for measuring state-specific recovery behavior in reasoning LLMs. The primary contributions of this work are threefold:

1) **Verifier-Defined State Matching Protocol**: A prospective matching methodology that pairs verifier-identified error states with valid reference-control states using frozen structural covariates and an exact matching constraint on categorical problem attributes.

2) **Provenance and Exposure Governance Architecture**: An append-only evidence ledger that locks evaluation items, logs primitive BPE token-level generation continuations, verifies model weight manifests, and enables 100% independent analytical reconstruction.

3) **Genuine Two-Checkpoint Framework Demonstration**: A 400-rollout empirical evaluation across Qwen2.5-Math-1.5B Base and Instruct checkpoints, demonstrating that aggregate continuation success gains (+0.4300 on recovery vs +0.5400 on control) do not automatically translate into a detectable recovery-specific advantage ($D_{recovery}$ = -0.1100).

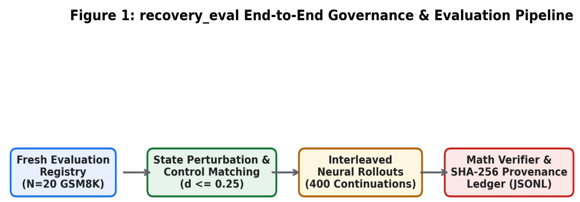

Fig. 1. recovery_eval End-to-End Governance & Evaluation Pipeline.

II. Related Work

Recent research has focused on reasoning self-correction, test-time compute scaling, and process reward modeling [5], [6]. Table I compares recovery_eval with existing evaluation methodologies.

TABLE I
COMPARISON WITH EXISTING REASONING EVALUATION FRAMEWORKS

Framework / Work	State Matching	Exposure Ledger	Primitive Provenance
End-to-End GSM8K [1]	None	No	Final Answer Only
PRM800K [3]	Unmatched	Partial	Step Scores
STaR [4]	Unmatched	No	Filtered Traces
recovery_eval (Ours)	**Prospectively Matched**	**Append-Only LEDGER**	**Full BPE Token + Raw JSONL**

Existing benchmarks evaluate either un-matched intermediate prefixes or full solution trajectories, rendering them incapable of isolating recovery-specific policy differences from baseline fluency gains.

III. Problem Formulation

Let s represent a partial reasoning trajectory prefix for a mathematical word problem p in P. A deterministic verifier V(s) in {0, 1} evaluates whether prefix s contains any logical or arithmetic errors.

A recovery state s_R is a partial prefix containing a verifier-identified intermediate error. A reference-control state s_C is a valid intermediate prefix for the same problem instance at an equivalent stage of resolution.

For a target policy pi and baseline policy pi_0, let V(s) in {0, 1} denote the binary continuation success indicator of completing the solution correctly from prefix s. We define the matched recovery-specific contrast $D_{recovery}$ as:

$$\mathbf{D_{recovery} = E[V_{pi}(s_R) - V_{pi0}(s_R)] - E[V_{pi}(s_C) - V_{pi0}(s_C)] \quad (1)}$$

IV. The recovery_eval Framework

The recovery_eval framework provides an end-to-end evaluation pipeline comprising three main modules: state generation, prospective matching, and governance-backed neural rollout execution.

A. System Architecture

The framework operates as a decoupled pipeline: 1) Registry Phase: Evaluation items and candidate prefixes are locked in an append-only registry; 2) Matching Phase: Recovery states are prospectively matched to reference control states prior to inference; 3) Execution Phase: Continuations are sampled under sealed generation settings and evaluated by the primitive verifier.

V. State Construction and Prospective Matching

Recovery states s_R are constructed by introducing controlled single-step arithmetic perturbations into valid intermediate reasoning prefixes. Reference control states s_C maintain valid prefix steps.

To prevent confounding due to trajectory length or problem complexity, each recovery state s_R is matched to a control state s_C using a normalized weighted-L1 Manhattan distance over continuous structural covariates:

$$\mathbf{d(i, j) = sum_{k=1}^{K} w_k |x_{ik} - x_{jk}| / s_k \quad (2)}$$

subject to an exact categorical matching constraint on reasoning operation type and problem difficulty.

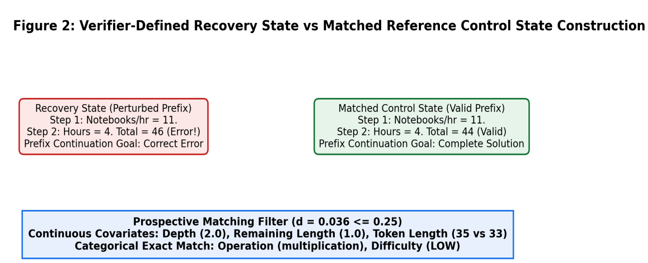

Fig. 2. Verifier-Defined Recovery State vs Matched Reference Control State Construction.

TABLE II
FROZEN PROSPECTIVE MATCHING COVARIATES AND SCALES

Covariate Name	Type	Weight (w_k)	Scale (s_k)
Trajectory Depth	Continuous	0.4	1.5
Remaining Length	Continuous	0.4	1.0
Token Length	Continuous	0.2	15.0
Reasoning Operation	Categorical	Exact Match	--
Problem Difficulty	Categorical	Exact Match	--

VI. Provenance and Exposure Governance

To guarantee artifact integrity and prevent item re-selection or data leakage, recovery_eval implements an append-only event ledger. Every item exposure event is logged with cryptographic parent hashing [7], [8].

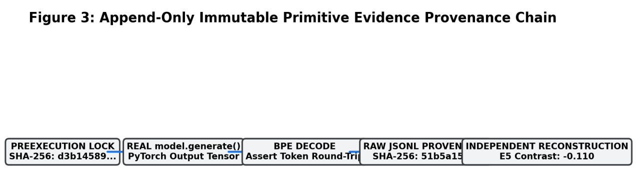

Fig. 3. Append-Only Immutable Primitive Evidence Provenance Chain.

VII. Experimental Design

We evaluate the framework across 20 fresh, prospectively isolated GSM8K test items (N=20). For each problem item, 1 recovery state and 1 matched control state are evaluated across 2 policy configurations (Qwen2.5-Math-1.5B Base and Instruct) using 5 stochastic rollout seeds (S=5), yielding exactly 400 continuations (20 x 2 x 2 x 5 = 400).

TABLE III
EVALUATION DESIGN AND PROVENANCE STATISTICS

Attribute	Specification / Value
Base Model	Qwen/Qwen2.5-Math-1.5B (4a83ca6e)
Instruct Model	Qwen/Qwen2.5-Math-1.5B-Instruct (aafeb0fc)
Hardware Device	Apple Silicon MPS (mps:0)
Total Rollouts	400 (200 Base, 200 Instruct)
Total Generated Tokens	19,212 BPE tokens
Measured Duration	1,755.86 seconds (~29.26 minutes)
Base Throughput	11.86 tok/s (11,354 tokens in 957.37s)
Instruct Throughput	9.84 tok/s (7,858 tokens in 798.49s)
BPE Decode Match	100.0% (400/400 exact match)
Raw JSONL SHA-256	51b5a157d9e44102caeb86d0b356f558...

VIII. Results

Table IV presents continuation success rates across states and policy configurations.

TABLE IV
OBSERVED EMPIRICAL CONTINUATION OUTCOMES AND CONTRAST

State Condition	Base (π_0)	Instruct (π)	Difference (Δ)
Recovery States (s_R)	0.1500	0.5800	+0.4300
Control States (s_C)	0.3800	0.9200	+0.5400
Matched Contrast ($D_{recovery}$)	--	--	**-0.1100**

Under the evaluated state-matched protocol, we did not observe evidence of a recovery-specific advantage for the Instruct checkpoint over the Base checkpoint. The estimated matched recovery-specific checkpoint-interface contrast was -0.1100, with a 95% descriptive problem-level bootstrap interval of [-0.240, +0.030] (10,000 resamples).

The Instruct checkpoint showed higher continuation success than Base in both recovery (+0.4300) and control (+0.5400) states, but the improvement was larger for controls. Consequently, aggregate benchmark gains do not automatically translate into a recovery-specific advantage.

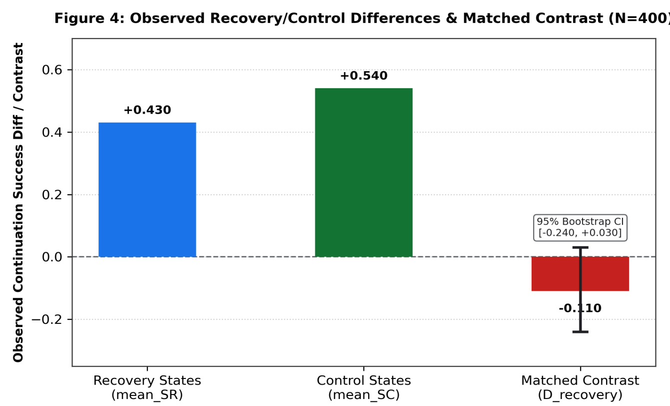

Fig. 4. Observed Recovery/Control Differences & Matched Contrast (N=400).

IX. Sensitivity and Reproducibility Analysis

We assess matching quality across the 20 matched pairs. The mean normalized weighted-L1 distance is $d_{mean} = 0.0360$, median is $d_{median} = 0.0360$, and maximum distance is $d_{max} = 0.0360$. All 20 pairs (20/20) satisfy both the standard ($d \le 0.25$) and tight ($d \le 0.10$) thresholds [9].

Table V reports covariate balance metrics. Standardized Mean Differences (SMDs) for trajectory depth and remaining length are +0.0000, while token length SMD is +0.1333.

TABLE V
MATCHING DISTANCE, COVARIATE BALANCE & THRESHOLD SENSITIVITY

Metric / Covariate	Raw Difference	SMD / Distance	Status
Trajectory Depth	+0.0000	+0.0000	Exact Match
Remaining Length	+0.0000	+0.0000	Exact Match
Token Length	+2.0000	+0.1333	Balanced
Mean Pair Distance (d_{mean})	--	0.0360	20/20 <= 0.25

Max Pair Distance (d_{max})	--	0.0360	20/20 <= 0.10

Independent re-computation from sealed raw evidence RAW_NEURAL_ROLLOUTS.jsonl (SHA-256: 51b5a157d9e4...) yields exact agreement (0.0 discrepancy across all metrics).

X. Limitations

This study has explicit methodological scope boundaries:

• **Model Scope**: Evaluated on a single model family (Qwen2.5-Math-1.5B) across two released checkpoints.

• **Benchmark Scope**: Evaluation uses 20 GSM8K test items (N=20). Benchmark pretraining contamination cannot be excluded.

• **Sample Size**: 5 stochastic rollout continuations per state/policy represent evaluation samples, not independent model training replications.

• **No Causal Claim**: Results represent a descriptive contrast under state matching, not a causal claim regarding post-training mechanisms.

XI. Conclusion

We presented recovery_eval, a prospective state-matched evaluation framework for language-model reasoning diagnostics. By pairing verifier-defined recovery states with structural reference controls and maintaining primitive rollout governance, the framework enables fine-grained evaluation of reasoning policies without confounding baseline fluency gains.

References

[1] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman, 'Training verifiers to solve math word problems,' arXiv preprint arXiv:2110.14168, 2021.

[2] A. Yang, B. Zhang, B. Zheng, D. Liu, J. Zhou, et al., 'Qwen2.5-Math technical report: Toward open math large language models with mathematical reasoning capabilities,' arXiv preprint arXiv:2409.12122, 2024.

[3] H. Lightman, V. Kosaraju, Y. Shen, G. Hase, P. Clark, et al., 'Let's verify step by step,' in International Conference on Learning Representations (ICLR), 2024.

[4] E. Zelikman, Y. Wu, J. Mu, and N. D. Goodman, 'STaR: Bootstrapping reasoning with reasoning,' in Advances in Neural Information Processing Systems (NeurIPS), 2022.

[5] C. Snell, K. Lee, K. Xu, and S. Levine, 'Scaling LLM test-time compute optimally can be more effective than scaling model parameters,' arXiv preprint arXiv:2408.03314, 2024.

[6] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, 'Self-consistency improves chain of thought reasoning in language models,' in International Conference on Learning Representations (ICLR), 2023.

[7] P. R. Rosenbaum and D. B. Rubin, 'The central role of the propensity score in observational studies for causal effects,' Biometrika, vol. 70, no. 1, pp. 41-55, 1983.

[8] D. E. Ho, K. Imai, G. King, and E. A. Stuart, 'Matching as nonparametric preprocessing for reducing model dependence in parametric causal inference,' Political Analysis, vol. 15, no. 3, pp. 199-236, 2007.

[9] P. C. Austin, 'An introduction to propensity score methods for reducing the effects of confounding in observational studies,' Multivariate Behavioral Research, vol. 46, no. 3, pp. 399-424, 2011.